NCVET Code
2022/ITES/NIELIT/06335

CONTACT DETAILS OF THE BODY SUBMITTING THE QUALIFICATION FILE

Name and address of submitting body:
National Institute of Electronics and Information Technology (NIELIT)
NIELIT Bhawan, Plot No 3, PSP Pocket, Institutional Area
Sector-8, Dwarka, South West Delhi, Delhi-110077

Name and contact details of individual dealing with the submission
Name: Dr. Manish Arora

Position in the organization: Joint Director (Academics)

Address if different from above: NIELIT Chandigarh

Tel number(s): 8264098112

E-mail address: manisharora@nielit.gov.in

List of documents submitted in support of the Qualifications File

Annexure I – Detailed Curriculum
Annexure II – Evidence of Need

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SUMMARY

1	**Qualification Title**	IT Application Engineer (B Level - IT)
2	**Qualification Code, if any**	**--**
3	**NCO code and occupation**	2522.0201, 2412.0200, 2412.0300, 2421.0101, 2431.0100, 2434.0101
4	**Nature and purpose of the qualification (Please specify whether qualification is short term or long term)**	Long term IT/ITES Skilling Course with the sole purpose to generate skilled manpower in the area of Information Technology (IT) at the national level by utilizing the facilities and infrastructure available with the institutions/organizations in the non-formal sector.
5	**Body/bodies which will award the qualification**	National Institute of Electronics and Information Technology (NIELIT) NIELIT Bhawan, Plot No 3, PSP Pocket, Institutional Area, Sector-8, Dwarka, South West Delhi, Delhi - 110077
6	**Body which will accredit providers to offer courses leading to the qualification**	National Institute of Electronics and Information Technology (NIELIT) NIELIT Bhawan, Plot No 3, PSP Pocket, Institutional Area, Sector-8, Dwarka, South West Delhi, Delhi - 110077
7	**Whether accreditation/ affiliation norms are already in place or not, if applicable (if yes, attach a copy)**	Yes, available at www.nielit.gov.in
8	**Occupation(s) to which the qualification gives access**	i. Freelancer (self-employed) ii. Full Stack Developer iii. Data Scientist/ Data Analyst iv. IoT Architect v. IoT Developer vi. Business Intelligence Analyst vii. Information Security Analyst viii. Training Faculty

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		ix. R & D Scientist x. Senior Software Developer xi. Consultant xii. Solution Architect
9	**Job description of the occupation**	After undergoing this qualification, candidates will be able to perform primarily following job roles. i. Freelancer (For self-employed) ii. Full Stack Developer iii. Data Scientist/Analyst iv. IoT Architect/Developer v. Business Intelligence Analyst vi. Information Security Analyst vii. Training Faculty viii. R & D Scientist ix. Senior Software Developer x. Consultant xi. Solution Architect
10	**Licensing requirements**	-NA-
11	**Statutory and Regulatory requirement of the relevant sector (documentary evidence to be provided)**	-NA-
12	**Level of the qualification in the NSQF**	**Level 6**
13	**Anticipated volume of training/learning required to complete the qualification**	2520 Theory: 660 Practical: 1140 Project: 600 ES: 120
14	**Indicative list of training tools required to deliver this qualification**	Given in Annexure - I
15	**Entry requirements and/or recommendations and minimum age**	NIELIT 'A' Level under DOEACC Scheme with graduation Or

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		Government recognized PGDCA/ BCA/ B.Sc. (IT)/ B.Sc. (Computer Science)
16	**Progression from the qualification (Please show Professional and academic progression)**	Academic: C-Level IT-> M.Tech (CS)->Ph.D Professional: - - Full Stack Developer/ Full Stack Web Developer-> Sr Full Stack Web Developer-> Project Manager - Data Scientist/Analyst-> Big Data Analytics Architect-> Analytics Manager -IoT Developer/ Architect -> Sr. IoT Developer/ Architect-> Chief Internet of Things Officer - Business Intelligence Analyst -> Business Intelligence Manager-> Business Intelligence Director - Information Security Analyst-> Information Security Engineer-> Security Architect, IT-> Director of IT Security -Training Faculty-> Sr Faculty-> Head of Dept - Freelancer (self- employed/ Entrepreneur)
17	**Arrangements for the Recognition of Prior learning (RPL)**	Candidates with qualification as mentioned at sr. 15 can register for examination directly
18	**International comparability** Where known (research Evidence to be provided)	NIELIT B and C Level Qualifications with three- and one-year experience, respectively are recognized as equivalent to Associate Certified IT Project Manager (CITPM) of National Infocomm Competency Centre (NICC) Singapore and Singapore Computer Society (SCS) as per the

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		MOU signed in August, 2002 between NIELIT and NICC & SCS. NIELIT A, B and C Level qualifications have been recognized as equivalent to Fundamental Information Technology Engineer, Software Design & Development Engineer and Application System Engineer Standards, respectively of Japan Information Technology Engineers Examination Center (JITEC) of Information Technology Promotion Agency
19	**Date of planned review of the Qualification**	25/08/2025

20		**Formal structure of the Qualification** **Mandatory components**			
		Title of Component and Identification Code/NOSs/Learning outcomes	**Estimated size (learning hours)**		**Level**
S.no	**Module code**		**Theory Learning Hours**	**Practical Learning Hours**	

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Semester I					
1	B0.1 R5	Basic Mathematics	45^	75^	5
2	B1.1- R5	Management Fundamentals and Information Systems	45	75	6
3	B1.2 –R5	Discrete Structure	45	75	6
4	B1.3- R5	Software Engineering	45	75	6
5	B1.4- R5	Operating System	45	75	6
6	B1.5- R5	DATA COMMUNICATIONS AND COMPUTER NETWORKS	45	75	6
Semester II					
7	B2.1- R5	Computer Based Statistical and Numerical Methods	45	75	6
8	B2.2- R5	Professional & Business Communication	45	75	6
9	B2.3- R5	Advanced Database Technologies	45	75	6
10	B2.4- R5	Computer Graphics and Multimedia Systems	45	75	6
11	B2.5- R5	Cloud Computing and Internet of Things (IoT)	45	75	6
Semester III					
12	B3.1- R5	Software Project Management	45	75	6
13	B3.2- R5	Artificial Intelligence and Machine Learning	45	75	6
14	B3.3- R5	Web Technologies	45	75	6

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15	-	Elective-1	45	75	6
16	-	Elective-2	45	75	6
Sub Total			660	1140	
24	B-PJ_R5	Major Project		600	6
25	-	Employability Skill		120	
		Total	2520		

^ Basic Mathematics is a bridge course not to be counted for the total duration of the course. Students from non-mathematics background will be taught Basic Mathematics Module.

List of Elective Modes are as below:

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1.	B3.E1 R5	Digital Marketing	45	75	6
2.	B3.E2 R5	-System Modeling & Computing Simulation	45	75	6
3.	B3.E3 R5	Distributed and Parallel Computing	45	75	6
4.	B3.E4 R5	Data Warehousing and Data Mining	45	75	6
5.	B3.E5 R5	Software Testing and Quality Assurance	45	75	6
6.	B3.E6 R5	Digital Image Processing	45	75	6
7.	B3.E7 R5	Accounting and Financial Management	45	75	6
8.	B3.E8 R5	Wireless and Mobile Communication	45	75	6
9.	B3.E9 R5	Blockchain Technology	45	75	6

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SECTION 1

ASSESSMENT

21	**Body/Bodies which will carry out assessment:** National Institute of Electronics and Information Technology (NIELIT) NIELIT Bhawan, Plot No 3, PSP Pocket, Institutional Area, Sector-8, Dwarka, South West Delhi -110077
22	**How will RPL assessment be managed and who will carry it out?** Yes, Given at Sr.no. 15.

23	**Describe the overall assessment strategy and specific arrangements which have been put in place to ensure that assessment is always valid, reliable and fair and show that these are in line with the requirements of the NSQF.** Theory examination for each module under the fifth revised syllabus would be for a duration of three hours and the total marks for each subject would be 100. Dates for the various activities related with examinations will be announced on NIELIT website, well in advance of the examinations Candidates are required to pass in all papers(Theory + Practicals) individually. ● All theory examinations will be conducted in the **OFFLINE (PEN- Paper)** mode. Each paper will consist of two parts. ● **Part I** carries 40 marks and contains forty questions, which is a combination of Objective type -Multiple choice questions, Fill-in the blanks, True/ False and Match the Following. ● **Part II** carries 60 marks and contains five subjective type questions. ● Project will be based on Specialized Areas **Pass Percentage** To qualify for a pass in a module, a candidate must have obtained at least 50% in each theory and practical examination each. The marks will be translated into Grades, while communicating results to the candidates. *The examination will be conducted as per the guidelines which are amended from time to time.

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ASSESSMENT EVIDENCE

24. Assessment evidences

B0.1-R5-Basic Mathematics

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Sequences & Series	After completion of this unit of module, the Learner will be able to i. Understand concepts of Sequences & Series
2. Matrices and Determinant	After completion of this unit of module, the Learner will be able to i. Understand Matrices & Determinant and solve System of Equations
3. Analytical Geometry	After completion of this unit of module, the Learner will be able to i. Understand Concepts of Coordinates and solve various problems related to it.
4. Vectors	After completion of this unit of module, the Learner will be able to i. Understand Concepts of Vectors and solve various problems related to Vector.
5. Differential Calculus	After completion of this unit of module, the Learner will be able to i. Understand Concepts of Functions, Limits and Continuity ii. Understand Concepts of Differential Calculus and solve various problems based on it.
6. Integral Calculus	After completion of this unit of module, the Learner will be able to i. Understand Concepts of Integral Calculus and solve various problems based on it.

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Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

B1.1-R5-Management Fundamentals and information Systems

Outcomes to be assessed/NOSs to be assessed	**Assessment criteria for the outcome**
1. Process of Management	After completion of this unit of module, the Learner will be able to understand the various processes involving in managing business. Further, learner will be able to learn how to plan, control and manage projects.
2. Information Systems Concepts	After completion of this unit of module, the Learner will be able to learn the commonly used information systems in any organization like financial, inventory etc.
3. Enterprise Systems	After completion of this unit of module, the Learner will be able to learn the enterprise level information system like ERP
4. System Development Process	After completion of this unit of module, the Learner will be able to learn the processes involved in developing the information systems at enterprise levels
5. Information Systems Security	After completion of this unit of module, the Learner will be able to learn the security risks in developing, deploying and implementing information systems. Further, the learner will be able to know the process to avoid or mitigate the risks.
6. Overview of Project Management	After completion of this unit of module, the Learner will be able to learn the basics of Project Management, its life cycle and different kind of management that are done during project execution.
7. Case Studies	After completion of this unit of the module, learner will be able to learn through case studies.

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Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B1.2-R5-Discrete Structure

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Sets & Functions	After completion of this unit of module, the Learner will be able to i. Understand concepts of Sets. ii. Implement Sets on given data.
2. Formal Logic	After completion of this unit of module, the Learner will be able to i. Understand and implement the concept of mathematical logic.
3. Analysis of Algorithms	After completion of this unit of module, the Learner will be able to i. Understand the mathematical foundation in analysis of algorithms. ii. Analyse the efficiency of algorithms using time and space complexity theory.
4. Storage and Feature Extraction of Graphs and fTrees	After completion of this unit of module, the Learner will be able to i. Understand the concept of storage and feature extraction of Graphs and Trees. ii. Implement various related algorithms.
5. Counting and Combinatorics	After completion of this unit of module, the Learner will be able to i. Understand fundamentals of combinatory. ii. Formulate and solve recurrence relations.
6. Dynamic Programming	After completion of this unit of module, the Learner will be able to i. Examine the different dynamic programs. ii. Apply the dynamic programs to solve any scenarios.
Means of assessment	Assessment details are given at Sr. no 23

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Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B1.3-R5- Software Engineering

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Software Engineering Fundamentals	After completion of this unit of module, the Learner will be able to understand the concept of the Software Engineering fundamentals
2. Software Requirements Analysis & Specification	Familiarize with the i. Software requirements specification standards, methods, and tools ii. Analysis Modelling using models -Data, Dataflow, Control Flow etc.
3. CASE diagrams and Tools	Practice on preparing i. User story on given scenarios ii. Use Cases using UML diagrams of [Structural diagrams & Behavioral diagrams
4. Software Design	Familiarize with i. Design Concepts and Principles, Design Process, Design Methods ii. Design thinking, functional design, prototype design iii. Theories on UI/UX, Dashboard iv. Design of documents-Business Requirements Document, Statement of work, High Level Design Document, Technical Detailed Design Specifications, Requirements Traceability Matrix
5. Development of Use cases and UML diagrams	Familiarize with i. Design Concepts and Principles, Design Process, Design Methods ii. Design documents-Business Requirements Document, Statement of work, High Level Design Document, Technical Detailed Design Specifications, Requirements Traceability Matrix
6. Coding and Software Testing	Familiarize the i. Coding standards ii. Testing driven design, Domain testing iii. Practice to Software Testing activities iv. Dynamic and pyramid testing v. Test result analysis

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7. Software Quality and Metrics	Familiarize the Software Quality Assurance, ISO9000 and 9001, CMM Levels and SIX sigma etc.
8. Software Maintenance and Configuration management	i. Can do hands-on Software Maintenance Process & Models, ii. Familiarize on the Configuration Management, Process
9. Object-Oriented Software Engineering	Familiarize with the OO Concepts and Approach, OO Analysis
10. Advanced Software Engineering Topics	Understand the concept of i. Advanced Software Engineering Topics Clean room approach and strategy, Component-based software engineering process, Reusability and Reengineering process. Agile and Scrum practices ii. Introduction to Software architecture: Architectural Styles, Pipes, and filters, MVC etc.
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

B1.4-R5- Operating System

Outcomes to be assessed/NOSs to be assessed	**Assessment criteria for the outcome**
1. Operating System Structures	i. Learn the OS structures
2. File System	i. Learn about the File System techniques in OS
3. Process Management	i. Learn about the concept of process. ii. Simulate communication between the processes.
4. Process Synchronization	i. Learn the synchronization concept among processes. ii. Simulate tools for synchronization iii. Understand Deadlock Handling and learn the algorithms

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5. Memory Management	i. Learn how OS manages memory ii. Learn how OS may improve the utilisation of the RAM iii. Understand Memory Management Schemes
6. Storage Management	i. Learn how mass storage and I/o are handled in OS ii. Understand I/O System Design and Interfaces iii. Work on Performance issues with I/O devices
7. Protection and Security	i. Learn about authorisation to operate in memory, CPU and files. ii. Controls an enforcement for security from unauthorised access, malicious destruction or alteration, and inconsistency
8. Virtualization	i. Learn about creating Virtual Machines ii. Learn types of VM's and their implementation iii. Create Virtual Machines
9. Distributed, Networks and Embedded OS	i. Know about Distributed , Network and Embedded OS ii. Learn their Network and Communication structure iii. Learn the Distributed File System and Remote File Access. iv. Know various Real Time OS
10. Case Study: Linux and Microsoft Windows	i. Learn the difference between Windows, and Linux Architecture ii. Learn Process Management and Scheduling iii. Learn about Memory and File Management iv. Learn the various commands of Linux and windows.
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B1.5-R5: Data Communications and Computer Networks

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
Introduction to Data Communications	After completion of this unit the candidate will be able to i. Various modes of Data communication ii. Components of data communications iii. Transmission modes
Introduction to Computer Networks	After completion of this unit the candidate will be able to i. Understand the concept of networking, various terminologies used in Networking ii. Understand various types of Networks, Network topologies iii. Network layers concepts and its merits and de-merits iv. Basics of OSI model and TCP-IP protocol suite
Physical Layer	After completion of this unit the candidate will be able to Understand how data travels physically and understand concepts of signals, modulation, multiplexing etc.
Data Link Layer	After completion of this unit the candidate will be able to i. Understand function of Data link layer, data framing, Addressing, error detecting codes ii. DLL-sublayers, Physical layer protocols iii. IEEE standards for wired and Wireless LAN
Network Layer	After completion of this unit the candidate will be able to Understand IP addressing (IPV4,IPV6) i. Understand Network layer protocols ii. Understand Routing Protocols
Transport Layer	After completion of this unit the candidate will be able to i. Understand function of transport layer and port addressing

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	ii. Understand Transport layer protocols. iii. Understand basics of congestion in network and various congestion control techniques
Application Layer	After completion of this unit the candidate will be able to Understand function of application layer and various protocols of this layer.
Multimedia in Data Communication	After completion of this unit the candidate will be able to Understand Multimedia in data communication and streaming protocols.
Communication Protocols in IoT	After completion of this unit the candidate will be able to : i. Explore data protocols for IoT ii. Explore Network Protocols for IoT.
Network Programming using Python	After completion of this unit the candidate will be able to : i. Work with TCP/UDP Sockets ii. Write Simple TCP/UDP Client and Server Program using Python iii. Handle Received Client Data over TCP Socket using Python. iv. Perform Operation on Data in Client-Server Program.
High Speed Networks	After completion of this unit the candidate will be able to Understand Packet Switching, Gigabit Ethernet, MPLS over IP , IP over SONET or SDH, IP over WDM , IP over DWDM, IP over fiber,GPON
QoS in Data Networks	After completion of this unit the candidate will be able to : i. Understand Integrated and Differential Services: ii. Understand protocols for QOS Support:
Introduction to Software Defined Networking	After completion of this unit the candidate will be able to: i. Understand SDN (Software Defined Networking). ii. Understand Software-Defined Wide Area Networking (SD-WAN) iii. Understand Network Function Virtualization (NFV).

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Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

B2.1-R5: Computer Based Statistical and Numerical Methods

Outcomes to be assessed/NOSs to be assessed	**Assessment criteria for the outcome**
1. Introduction to Statistics & Probability	After completion of this unit of module, the Learner will be able to : i. Understand basic concepts of probability & Statistics ii. Solve basic Statistical problems.
2. Distribution Functions	After completion of this unit of module, the Learner will be able to Understand and implement discrete and continuous distribution functions like Binomial, Poisson and Normal.
3. Statistical Inference	After completion of this unit of module, the Learner will be able to i. Understand the concepts of Testing of Hypothesis and its various terms. ii. Test various hypotheses on given data.
4. Regression Analysis	After completion of this unit of module, the Learner will be able to Understand concept of different types of Regression Analysis and its implementation.
5. Computer Arithmetic and Errors	After completion of this unit of module, the Learner will be able to Understand and implement the concept of Number representation and errors.
6. Algebraic and Transcendental Equations	After completion of this unit of module, the Learner will be able to

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	i. Understand Algebraic and Transcendental Equation, ii. Implement Bisection Method, Iteration Method, and Newton – Rap son Method.
7. System of Linear Equations	After completion of this unit of module, the Learner will be able to i. Understand various concepts in System of Linear Equations. ii. Solve System of Linear Equations using different methods.
8. Numerical differentiation & Integration	After completion of this unit of module, the Learner will be able to i. Understand various concepts in numerical differentiation and integration. ii. Apply different techniques related ND and NI.
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.2-R5: Professional & Business Communication

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Basics of Communication	After completion of this unit of module, the Learner will be able to i. Understand the need of good communication skill ii. Learn Basics of Communication
2. Communication Techniques	After completion of this unit of module, the Learner will be able to learn i. Speaking skills ii. Negotiating Skills, Etiquettes iii. Comprehension Skills
3. Listening Practices	After completion of this unit of module, the Learner will be able to know i. Importance of listening ii. Good Body Language iii. Effective Listening
4. Communication in Groups and Teams	After completion of this unit of module, the Learner will be able to i. Performing Effectively in Teams; ii. Communication in meetings
5. Technical Communication	After completion of this unit of module, the Learner will be able to i. Write Technical Reports ii. Write letters, Memos, emails etc.
6. Interpersonal Communication Skills	After completion of this unit of module, the Learner will be able to i. Build Interpersonal Skills in the Workplace. ii. Communicate in official environment
7. Resumes and Interviews	After completion of this unit of module, the Learner will be able to i. Write resume, cover letters etc. ii. Learn how to face Employment and Appraisal interviews etc.

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8. Presentation Skills	After completion of this unit of module, the Learner will be able to learn i. Facets of Presentation ii. Extempore Speech, Public Speaking etc.
9. Technology in Communication	After completion of this unit of module, the Learner will be able to learn i. Role of communication technology in office. ii. Concept of virtual office
10. Collaborative Technology	After completion of this unit of module, the Learner will be able to i. Use groupware ii. Learn Collaborative Computing
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.3-R5: Advanced Database Technologies

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Recap1 on Relational model and relational algebra	After completion of this unit, the Learner will be able to understands the concept of Relational model and relational algebra, and related topics
2. Recap2 on E-R models and Normalization	To Familiarize with the E-R models and Normalization process etc
3. Database Architectures and Open source databases	To Understand the application Architectures, Database Schema, Concept of Open source databases such as Maria DB and Mongo DB, Database-as-a-service (DBaaS);
4. Database Properties and Database securities	Familiarize with Database Properties, Database Consistency, Constraints and Data Integrity, Database Transactions, The concept of Database securities, Common threats, Security Levels, some possible solutions
5. Query languages-SQL/Non-SQLs	Develop use cases using the Procedural or Non-Procedural Query Language /SQLs/Non-SQLs
6. Transactions and Concurrency	Familiarize with the Transactions and Schedule, Concurrency Control and Develop use cases using Queries
7. Durability	Familiarize with the Durability process and Checkpoint facilities and Develop use cases using Queries
8. Recovery manager	Familiarize with the Recovery manager and its process and Developed use cases using Queries
9. Data Warehouses and OLAP	Familiarize with the Steps to develop a Data Warehouse and ETL process
10. Advance topics	Understand the advanced Topic-Distributed Database, Object-oriented databases, Temporal databases, Image, multimedia, and spatial databases and Indexing techniques
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.4-R5: Computer Graphics and Multimedia Systems

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Graphics Hardware – Primitives	After completion of this unit of module, the Learner will be able to Understand the basic hardware primitives used in Computer Graphics.
2. Basic Mathematical Concepts for Computer Graphics	After completion of this unit of module, the Learner will be able to Understand the Mathematics behind the implementation of Computer Graphics
3. Graphics Operations – Clipping, Filling	After completion of this unit of module, the Learner will be able to Understand the various algorithms for clipping and filling of various shapes.
4. Transformation 2D – 3D & Projections	After completion of this unit of module, the Learner will be able to Understand the transformations of 2D and 3D. Further, learner will learn the Parallel, Perspective and Isometric projections are illustrated in this unit.
5. Object Representation	After completion of this unit of module, the Learner will be able to i. Understand the representation of various surfaces. ii. Understand the Hermit curve, Bezier curve and B-spine curve. iii. Understand the Shading Algorithms.
6. Multimedia Systems	After completion of this unit of module, the Learner will be able to i. Learn Basics of Multimedia. ii. Basic editing in Image. Editing Software. iii. Concepts of Audio. Basic Video Editing.
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.5-R5: Cloud Computing and Internet of Things (IoT)

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Cloud Computing	After completion of this unit of module, the Learner will be able to understand i. Cloud Computing Architecture, Models ii. Virtualization iii. Security aspects
2. IoT Hardware Platform, Sensors & Actuators	After completion of this unit of module, the Learner will be able to i. Hardware platform for IoT ii. Low power microcontrollers-ARM cortex M Series iii. Hardware for prototyping IoT Gateway
3. IoT Connectivity & Cloud Platform	After completion of this unit of module, the Learner will be able to i. IoT connectivity standards ii. IoT layered architecture & Protocols iii. Network Topologies iv. IoT Cloud Platform
4. IoT Framework	After completion of this unit of module, the Learner will be able to i. IoT architecture reference model ii. Identity and Access management iii. IoT open-source framework, commercial framework
5. Use Cases – IoT & Cloud Services	After completion of this unit of module, the Learner will be able to IoT Case studies in Agriculture, Healthcare, Manufacturing
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.1-R5: Software Project Management

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. The recap on Software Project and Introduction to Project Management	After completion of this unit, the Learner will be able to i. Understand the concept of Software Engineering, its Process and Models etc. ii. The concept of a project, its principles, project life cycle, and CMM model
2. Project Management Issues	Familiarize with the Basic project management functions and forces of a project. Product and service management, Knowledge management
3. Project Planning and Project Estimation	Practice the Project Planning and Project Estimation using the available methods and algorithms models and tools
4. Project Scheduling	Familiarize with Project Scheduling activities, sequencing using available techniques and tools
5. Risk Analysis and management	Can do the identification of risk, assessment and tracking for better Risk Mitigation and Management
6. Software measures and metrics	Practice to measure the software details using available metrics
7. Quality Management	Can do hands-on practice on quality management of a project using the models and can generate review reports- defect and quality etc.
8. Project Tracking Techniques and Monitoring	Can do hands-on practice on Tracking and Monitoring of a project using available Chart and graphical tools
9. Configuration Management, and Project Closure	Familiarize with the activities in the Configuration Management and planning, Change management and Version Control System etc. and familiarize with Defect Collection, Audit and Project Closure analysis
10. Agile Management	Understand the concept of Agile Management-Team members and Leadership together with Scrum practices
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.2-R5: Artificial Intelligence and Machine Learning

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction to Artificial Intelligence	i. Describe the building blocks of AI Systems. ii. List the environment and goals of agent-based systems and draw the design of an Agent.
2. Applications of AI	Identify the suitability of applying AI as a solution, based on context of applications.
3. Advanced Python (Numpy, Pandas, Matplotlib)	i. Understand Numpyndarray and various functions of numpy ii. Importing Data from various sources (csv, txt, excel) iii. Reading and manipulate csv files iv. Use Pandas for data science, Performing EDA, Data Wrangling, Combining Datasets-Merge and Join, Grouping Data. v. Make various types of Graphs and Plots using Python Graphical libraries.
4. Machine Learning	i. Machine Learning Introduction, Supervised and Unsupervised Learning. ii. Tasks performed by Machine Learning Algorithms – Classification, Regression, Clustering iii. Concept of Training and Testing Data, Splitting the data using python modules iv. Solving a Regression Problem using Python: Data Preparation, Apply Machine Learning Algorithm: Linear Regression, Decision Tree Regress or etc. , use of Sklearn package v. Solving a Classification Problem using Python: Data Preparation, Apply Machine Learning Algorithm: Logistic Regression, Decision Tree Classifier, K-Nearest Neighbour Classifier etc.

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	vi. Understanding Mathematics behind Machine Learning Algorithm vii. Solving a Clustering problem by applying K-Means Algorithm in Python. viii. Performance evaluation metrics of machine learning algorithms-Accuracy Score, Confusion Matrix, Root Mean Squared Error etc. ix. Over fitting, under fitting. x. Cross Validation, Optimizing Model Hyper parameters.
5. Deep Learning	i. Learn basics of deep learning concepts. ii. Learn to train and implement deep learning algorithms. iii. Understand the implementation of AI through real world examples of images. iv. Learn neural network using tensor flow. v. Making Deep Learning model using Image Data: Digit Recognition Model.
6. Computer Vision	i. Introduction to Computer Vision ii. Image Representation and Analysis iii. Write programs using machine learning algorithms in Open CV for detection and recognition of images. iv. Learn Face recognizers. v. Training image data and prediction.
7. Natural Language Processing	i. Understand the basics of text processing. ii. Working with Text Data, Count Victimization, and sparse matrix. iii. Learn application of natural language processing. iv. Understand the concepts of NLP and process linguistic data using the popular algorithms.
8. Case Study	Case Study of use of Machine Learning in Real Life Applications
Means of assessment	Assessment details are given at Sr. no 23

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Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

B3.3-R5- Web Technologies

Outcomes to be assessed/NOSs to be assessed	**Assessment criteria for the outcome**
1. Introduction to Java and OOPs concepts	After completion of this unit of module, the candidate will be familiar to i. OOPs Concept ii. Java Execution Model
2. Java Concept	After completion of this unit, candidate will have in depth knowledge of i. Use of variables, literals, data types and arrays. ii. Working with each data item iii. Java operations, loops, conditional & jump statements
3. Database Concept and JDBC Connectivity	After completion of this unit, candidate will be will be familiar with the i. JDBC Architecture ii. JDBC APIs that provide various classes and interfaces to develop a JDBC application using SQL Transactions
4. Servlet API	After learning this unit, candidate will be able to: i. Understand role of Java Servlets in the overall Java 2 Enterprise Edition architecture, and as the best Java solution to HTTP application development ii. Use request and response objects provided to a servlet to read parameters and to produce an HTML response iii. Develop interactive web applications using HTML forms and servlets
5. JSP and database connectivity	After learning this unit, candidate will be able to: i. Design and build robust and maintainable web applications

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	ii. Create dynamic HTML content with Servlets and Java Server Pages, using the JSP Standard Tag Library (JSTL) iii. Use JSTL and other Custom Tag Libraries to separate Java and HTML code iv. Access databases with JDBC
6. Hibernate Framework	After completion of this unit, candidate will have in depth knowledge of i. Understand the concepts of object/relational mappings ii. Create Hibernate mappings iii. Retrieve and update persistent objects using Hibernate iv. Use HQL (Hibernate Query Language) and criteria queries
7. Java Web Frameworks: Spring MVC	After completion of this unit, candidate will be able to i. Use the Spring MVC web framework and form tag library to develop flexible web applications ii. Understand the need for the Spring framework iii. Use inversion of control to increase flexibility and testability of applications iv. Integrate Spring with the Hibernate ORM framework
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E1-R5- Digital Marketing

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Fundamentals of Digital Marketing	i. Understand the concept of marketing the products and services digitally ii. Understand the definition and origin of Digital Marketing iii. Identify the benefits and difference of Digital Marketing vs. Traditional Marketing iv. Describe the phases and approaches in Digital Marketing
2. Search Engine Optimization (SEO)	i. Understand the concept of Search Engine Optimization ii. Describe the types of SEO iii. Understanding SEO Myth busters iv. Optimizing the site structure
3. Search Engine Marketing (SEM)	i. Understand the concept and importance of Search Engine Marketing ii. Learn Ad writing techniques ii. Identify and strategizing PPC (Pay-per-click) campaigns v. Learn Link Building
4. Social Media Marketing (SMM)	i. Learn the concept and importance of Social Media Marketing (SMM) ii. Building successful Social Media Strategy iii. Setting SMM goals and bookmarking iv. Marketing on Social Media platforms such as Facebook, LinkedIn and Instagram v. Understanding Trends in Digital Marketing and Advertising

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5. Content and E-mail Marketing	i. Learning Content writing techniques ii. Understanding the concept of Blogging, creating effective Blogs and Posts iii. Building blogging strategies powered by SEO iv. Micro-blogging using Koo and Twitter v. Creating Vlogs, Uploading on media streaming websites vi. Learning fundamentals and best practices of e-mail marketing vii. Developing e-mail strategies viii. Building and importing e-mail lists ix. Using segmentation strategy and segmentation lists
6.Affiliate Marketing	i. Understanding concept and importance of Affiliate Marketing ii. Understanding the difference between referral and Affiliate Marketing ii. Accessing and using Affiliate Marketing platforms
7. Web Analytics	i. Learning Web Analytics ii. Dashboard Preview& Customization iii. Real-Time Report iv. Audience Demographics v. UTM Parameters vi. Social Referrals and Referral Traffic vii. Google Search Console viii. Campaigns, Analysing Landing & Exit Pages ix. Event Tracking
8. Keyword Planning and Social Media Ad Campaigns	i. Define Keyword planning and its importance ii. Keyword Research iii. Learning Keyword Planning Tools, Goals and bidding Creating and running Social Media Ad Campaigns iv. Boosting techniques on social media channels v. Using Social Media Ad Campaign analytics vi. Online Reputation Management (ORM)
9. Lead Generation and Marketing Automation	i. Understanding Lead generation strategy ii. Capturing leads from sources ii. Understanding Marketing automation tools v. Setting up email and SMS auto responders, Virtual Assistants

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Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E2-R5- System Modelling & Computing Simulation

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction	i. Basics of simulations ii. Advantages & disadvantages
2. Statistical Models in Simulation	i. Able to model deterministic systems and differentiate between nonlinear and linear models. ii. Understand the numerically simulate ordinary differential equations and deterministic systems.
3. Random-Number Generation	i. Numerical Techniques Sources and Propagation of Error
4. Input Modelling	Understanding of building model
5. Estimation of Absolute Performance	Able to estimate and validate a model based upon input and output data.
6. Measures of performance and their estimation 7. (Verification, Calibration and Validation)	Be able to model and simulate stochastic and discrete event systems. Able to estimate and validate a model based upon input and output data.
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E3-R5- Distributed and Parallel Computing

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction to Distributed and Parallel Computing Technologies	After completion of this unit of module, the Learner will be able to . Know the basics of Distributed and Parallel Computing technologies. . Know the benefits of Distributed and Parallel Computing technologies.
2. Distributed Computing	After completion of this unit of module, the Learner will be able to ● Know the evolution of Distributed systems ● Know the Distributed File systems (DFS) ● Understand the role of transaction and concurrency controls in Distributed Systems ● Gain knowledge of mutual exclusion algorithms ● Know the basics of edge computing ● Know the Challenges in Federating Edge Resources ● Know the method Data Management in Edge Computing
3. Parallel Computing	After completion of this unit of module, the Learner will be able to ● Know the parallel computing and its architecture ● Know working of OPENMP ● Know PRAM and its model
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E4-R5- Data Warehousing and Data Mining

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction	After completion of this unit, the Learner will be able to uunderstand the basics concept of Dara warehouse and Data mining
2. Data Warehousing	Familiarize with the Data warehouse, multidimensional database and Building steps of a Data Warehouse etc.
3. ETL processes	Implement the building of warehouse and possible process of ETL
4. Data Mining Primitives	Familiarize with the basis of data mining and Knowledge Discovery Database
5. OLAP engines and OLAP operations	Familiarize with the OLAP Engines and Develop use cases on Data Modeling using OLAP operations
6. Supervised and un-supervised techniques	Development of applications using Supervised techniques -Classification techniques
7. Association algorithms	Development of applications using Association Analysis, A priori algorithme etc.
8. Un-supervised algorithms	Development of applications using Un-supervised algorithms- Clustering Techniques etc.
9. Advanced Mining Concepts	Familiarize with the Data Mining attributes, and other topics- object oriented databases, spatial databases and multimedia databases, time series databases, Graph mining
10. Applications	Understand the advanced Topic- i. Exploration of related applications. ii. Mining-Pattern, opinion, Social media and crowd-sourcing data mining iii. Uses of data mining packages iv. Web content Mining, Bioinformatics v. Temporal and Spatial Data Mining:
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E5-R5- Software Testing and Quality Assurance

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction	After completion of this unit the Learner will be able to understand the concept of Software testing and quality assurance practices
2. Industry practices and Software Quality	Familiarize with the Industry practices and Software Quality
3. Understanding Pre-testing documents	Understands the Pre- testing documents and its purposes
4. Testing Techniques and Strategies	i. Familiarize with Testing Terminologies, activities in Testing and designing the Test ii. Practice the Unit Testing process iii. Practice the other Integration, system, acceptance testing etc.
5. Building Test Cases and Plans	Developed Test Cases, Plans, and require reports
6. Verification, Validation and Debugging Technique and Tools	Familiarize with the Requirement verification, Coding standards, Walk through etc. and Debugging Technique and Tools, debugging, tracing
7. Quality Assurance and Standards	Familiarize the Software Quality Assurance and related standards
8. Post testing activities and documents	Understands the Post testing documents and its purposes
9. Agile Software development practices and Agile Testing	Familiarize with advance testing topics- Agile testing, test pyramid etc.
10. Version Control	Understand the practices of version Management, Changed management
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3. E6-R5- Digital Image Processing

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction	After completion of this unit, the Learner will be able to understand the basics concept of Image processing and its applications
2. Image Digitization	Familiarize with the Image Digitization process.
3. Image Enhancement	Implement the process of Image enhancement techniques
4. Image Restoration	Implement the process of Image Restoration techniques
5. Image Compression	Implement the process of Image Compression techniques
6. Image Segmentation	Implement the process of Image Segmentation techniques
7. Image registration and Multi-valued Image Processing	Familiarize with concept on Image registration and Multi-valued Image Processing
8. Image Transformations	Implement the process of Image Transformations
9. Object detection process	Implement the process of Object detection process
10. Introduction to Virtual Reality and Augmented Reality	Understand the advanced Topic-Introduction to Virtual Reality and Augmented Reality
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E7-R5- Accounting and Financial Management

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Basic Accounting System	After completion of this unit of module, the Learner will be able to understand Accounting and its Functions, Scope of Accounting, Emerging Role of Accounting, Accounting as an Information System, Role and Activities of an Accountant, Accounting Personnel, Nature of Accounting Function, Organization Chart for Accounting and Finance
2. Accounting Concepts and Standards	After completion of this unit of module, the Learner will be able to understand Accounting Framework Accounting Concepts, Accounting Standards, Changing Nature of Generally Accepted, Accounting Principles (GAAP),Attempts towards Standardization, Accounting Standards in India
3. Preparation of Profit & Loss & Balance Sheet	After completion of this unit of module, the Learner will be able to understand i. Accounting Equation ii. Classification of Accounts Definitions of Journal and Ledger

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	iii. Journalizing Process Ledger Posting iv. Balancing an Account v. Trial Balance vi. Objectives of Preparing Trial Balance vii. Preparation of Profit & Loss and Balance Sheet. Ledger Posting and Trial Balance. viii. Use of any Financial Accounting software either Licensed or Open source
4. Financial Management	After completion of this unit of module, the Learner will be able to understand, Cash and Treasury Management, Treasury Risk `Management, Functions of Treasury Department, Facets of Cash Management, Motives for Holding Cash, Cash Planning, Determining the Optimum Cash Balance, Methods of Cash Flow Budgeting, Investing Surplus Cash Collection and Disbursements. Budgeting and Budgetary Control.
5. Inventory & Receivables Management	After completion of this unit of module, the Learner will be able to understand, Reasons for Holding Inventory, Objectives of Inventory Management, Techniques of Inventory Control. Various techniques of receivable management

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6. Portfolio Management and IT applications	After completion of this unit of module, the Learner will be able to understand, what is portfolio management, Portfolio management methods Design and implementation of portfolio management, Risk Management, Disaster Management, Portfolio management issues and challenges, Tools and techniques
7. Social, Ethical and Legal Aspects	After completion of this unit of module, the Learner will be able to understand Society in information age, Moral dimensions and information age, Technology trends and ethical issues, Ethical principal and dilemma, Responsibility, accountability and liability Information right and acts
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E8-R5- Wireless and Mobile Communication

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Wireless Communication Fundamentals	i. Radio frequency fundamentals ii. Modulation Techniques for Wireless and Mobile communication. iii. Wireless Frequency Spectrum
2. Cellular & Mobile technologies	Basic cellular system concept Cellular standards and generations
3. Wireless LAN Technologies	i. Wireless LAN overview ii. WiFi standards iii. WiFi security
4. WSN and WPAN Technologies	i. Wireless Sensor Network basics ii. WSN Topology iii. WSN standards iv. WPAN standards & Applications
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

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B3.E9-R5- Blockchain Technology

Outcomes to be assessed/NOSs to be assessed	Assessment criteria for the outcome
1. Introduction	After completion of this unit, the Learner will be able to understand the objectives, and importance of the course
2. Fundamentals of Blockchain Technology	Familiarize with Properties of Blockchain, Types of Systems, possible application areas and concept of the smart contracts, etc.
3. Consensus Mechanism	Familiarize with the consensus Mechanisms, Proof-of-Work, Mining and Incentives, Proof-of-Authority, Proof-of-Stake, Practical Byzantine Fault Tolerance (PBFT),etc
4. Encryption & Decryption Algorithm	Implement of Encryption & Decryption algorithms-RSA, Ceaser Cipher, Rijndael algorithm logic, Message Authentication Code
5. Crypto currencies and Bitcoin	Familiarize with crypto currencies, crypto currencies, Digital cash, Wallets, and Altcoins, etc
6. Types of Blockchain	Development of applications using Public Blockchain: (Ethereal), Ethereal Virtual Machine (EVM),
7. Private Blockchain	Development of applications using Hyperledger, uPort, SlockIt, Crypto-kitties, Augur, etc.
8. Implementation/Case studies	Development of

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	i. Private application son the Ethereal Blockchain platform. ii. Smart Contracts, iii. DApp Development environment
9. Standards and possible attacks	Familiarize with the Development Tools and libraries, Front End Technologies, Attacks on Smart Contracts, etc.
10. Use Case	Developing a use case and implementation as a mini project
Means of assessment	Assessment details are given at Sr. no 23
Pass/Fail	Pass/Qualifying Criteria is also given at Sr. no 23.

***Assessment for the Qualification will be conducted as per the guidelines as applicable from time to time.**

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SECTION 2

25. EVIDENCE OF LEVEL

Title / Name of Qualification/Components: A Level			**Level: 5**
NSQF Domain	**Outcomes of the Qualification/Component**	**How the outcomes relates to the NSQF level descriptors**	**NSQF Level**
Process	Job that requires well developed skill, with clear choice of procedures in familiar context. Requires wide range of specialized theoretical & Practical skills involving variable routine and non-routine context. Assess how different area of IT in the organization work together. Develop, maintain and upgrade software. Monitor performance and usage of available resources.	Demands a wide range of specialized technical skill, clarity of knowledge and practice in broad range of activities involving standard and non-standard practices. The individual in this role needs to have thorough understanding of IT processes and how to satisfy the requirement of the organization's clients /customers. Individuals will be expected to guide team members to carry out all assigned tasks safely and securely.	**6**
Professional Knowledge	Individual will have factual and theoretical knowledge in broad contexts within a field of work. Full resource utilization, skill Development and skill to update Knowledge time to time. Explain and Demonstrate Professional knowledge includes software setup, system recovery, back-up, upgradation & creating deployment plans.	Individual will carry out a broad range of work which requires wide ranging specialized theoretical and practical skills within a field of work or study. The individual will have in-depth knowledge of the different fields of IT and be able to guide junior associate to achieve the same.	**6**

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Professional Skills	They plan tests, prepare tests cases, generate test data and perform testing on test data to generate solutions to specific problems. They have good communication Skill and creative ideas for unique design of product.	The individual should have the ability to use various s/w in different platform & to monitor the performance and usage of resources. They should also know how to generate solutions to specific problem in the field of work. Individual is also expected to monitor and guide/ aid staff to complete their task in more effective manner and will be responsible for resolving day to day operational problems.	**6**
Core Skills	They should be good in communication, understanding, collecting information and logical skills. They can give new ideas for developments. They can identify system performance metrics and guide processes that track and measure performance of deployed systems.	Individual is expected to be able to communicate clearly in speech and writing and may be required to apply mathematical processes & expected to understand capability of team member and accordingly allocate work and check on the progress of the same, they should be able to guide/ lead teams on work within their capability.	**6**

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Responsibility	They have full responsibility for output of group and self-development. They can design/ upgrade/maintain websites, database, software and provide solution in a conducive & Non-Conductive environment. Check-up procedures to ensure that project objectives are finished within specified time frames. Apply balanced judgments to different situations, Follow rule-based decision-making processes & complete accurate well written work. Make decisions on suitable courses, convey relevant information to teams Manage performance of teams and motivate them to their full potential.	Individuals will be responsible for the completion of their own work and they will have some responsibility for other's work & the team learning process. Individual will be able to make choices about the best procedure to adopt to address problems where the choices are clear and they will require well developed practical and cognitive skills to complete their work.	**6**

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SECTION 3

EVIDENCE OF NEED

26	**What evidence is there that the qualification is needed? What is the estimated uptake of this qualification and what is the basis of this estimate?**	
	Basis	**In case of other Awarding Bodies (Institutes under Central Ministries and states departments)**
	Need of the qualification	This Qualification is designed to meet the increasing manpower requirements in computers after discussion with various ministries- MHRD, Industries etc. The 'A' Level of NIELIT is recognized as equivalent to Advance Diploma Level Course by the Government of India for the purpose of employment vide Notification No. F.18-23/92-TD.V/TS dated 1st March, 1995 and F.18-23/92-TD.V/TS-IV dated 10th April, 1996 issued by Ministry of Human Resources Development.
	Industry Relevance	The job role defined for the qualification is as per the National Qualification of Occupation 2015 which is developed by Employment Directorate under the ministry of Labour and Employment in collaboration with different industry partners and as per ILO guidelines. This justifies the qualification is very much relevant for industry.
	Usage of the qualification	With this qualification, candidates will be equipped with required knowledge to develop software. Candidates can be employed as Full Stack Developer/ Data Scientist/ Analyst/ IoT Architect/ IoT Developer/ Business Intelligence depending on the specialization.
	Estimated uptake	NIELIT is having 43 Centres and 140+ accredited Centres spread all over India and approximate 1000 candidates per year are expected to register in this course.

27	**Recommendation from the concerned Line Ministry of the Government/Regulatory Body. To be supported by documentary evidences** This qualification is run by National Institute of Electronics & Information Technology (NIELIT). It is an Autonomous Scientific Society under the

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	administrative control of Ministry of Electronics & Information Technology (MoE&IT), Government of India.
28	**What steps were taken to ensure that the qualification(s) does (do) not duplicate already existing or planned qualifications in the NSQF? Give justification for presenting a duplicate qualification.** As the understanding and adoption models of QPs evolve in the industry and across its sub-sectors, we foresee consolidation of qualification packs as a natural progression. The Qualification does not exist as per information available in the public domain.
29	**What arrangements are in place to monitor and review the qualification(s)? What data will be used and at what point will the qualification(s) be revised or updated? Specify the review process here** The Qualification is monitored and reviewed normally in 3 years. The following data will be used 1. Results of assessments 2. Employer feedback will be sought post-placement 3. Student feedbacks 4. Workshops and seminar for reviewing the qualifications 5. Industry Requirements 6. Consultation/ Tie-up with Industries or Expert for review of the Curriculum

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SECTION 4
EVIDENCE OF PROGRESSION

30	**What steps have been taken in the design of this or other qualifications to ensure that there is a clear path to other qualifications in this sector?** ***Show the career map here to reflect the clear progression*** This qualification consists of both technical and analytical skills and designed in such a way that it will provide core concepts of IT Skills and leading various programming techniques. The candidate can choose a field of his interest from the list of electives and proceed further in that particular field to pursue advance course or one can work in the specialized filed. This course gives link to higher qualification which is existing like B level of NIELIT A successful candidate will get a certificate from NIELIT, which gives the following options of progression to the candidate i. Freelancer (self-employed) ii. Full Stack Developer iii. Data Scientist/Analyst iv. IoT Architect v. IoT Developer vi. Business Intelligence Analyst vii. Information Security Analyst viii. Training Faculty

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